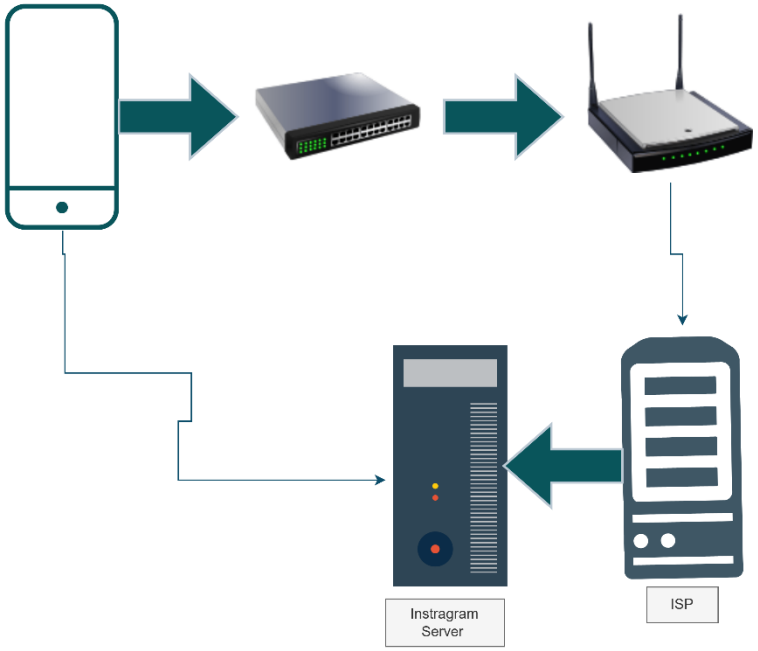
1) Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2) Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Comparison of Three WAN Technologies

WAN Technology	Speed	Cost	Real-Life Usage Example
1. MPLS (Multiprotocol Label Switching)	High and predictable; depends on the service plan	High	Large companies like Zomato may use private WAN connectivity to connect offices, data centers, and branches.
2. Leased Line	Dedicated, symmetrical bandwidth; often highly reliable	High	Banks, large offices, and critical systems such as railway ticketing infrastructure may use dedicated connectivity.
3. Broadband Internet	Medium to very high; speed varies by plan and network congestion	Low to medium	Home Wi-Fi, small offices, shops, and restaurants using internet services.

**3) Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.**

1. Connect to public Wi-Fi: Your phone or laptop connects to the coffee shop's Wi-Fi router.

2. Start the VPN: The VPN app establishes an encrypted tunnel between your device and a VPN server over the public WAN.

3. Send data securely: When you open YouTube or Spotify, your internet traffic travels through the encrypted tunnel to the VPN server.

4. Reach the app's servers: The VPN server forwards your request to YouTube or Spotify and relays the response back through the tunnel.

4) Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Two reasons:

1. Reliable and efficient connectivity: MPLS provides predictable network performance, helping warehouses update inventory, process orders, and synchronize data efficiently.
2. Secure branch connectivity: It supports managed connectivity between warehouse networks, helping centralize communication and apply consistent network policies.